Repairing Weak Seed Prompts with Iterative Teacher Feedback for Frozen Local Moral Judgment

Anonymous Submission Draft

Abstract

We study a narrow but practically important prompt-only question: can a strong teacher iteratively repair a relatively weak frozen seed prompt for a frozen small local student? We focus on ETHICS commonsense moral judgment, a local Qwen 0.5B-class student, and Gemini as the teacher. The central methodological choice is to optimize against a frozen seed prompt rather than demanding that the teacher beat an already-strong expert-written prompt outright. This makes the primary claim sharper, fairer, and easier to evaluate under paired testing.

The paper-facing protocol uses nested cross-fitting, deterministic local-only student inference, a fair one-shot full-budget teacher control, structured prompt edits, and transfer-check gating. The main comparison is `iterative_from_seed` versus `seed_frozen`; the expert prompt is benchmark-only and remains outside the teacher loop. The repository package adds per-run reporting, duplicate-group bootstrap confidence intervals, compact paper assets, and a release bundle that carries the exact report, paper, configs, prompts, figures, tables, and teacher-manifest summary used for review.

1 What We Claim and What We Do Not Claim

What we claim. A strong teacher can iteratively repair a relatively weak frozen seed prompt for a frozen local student under prompt-only constraints and fair controls.

What we do not claim. We do not claim broad moral reasoning improvement, domination of the strongest human-written expert prompt, or pristine confirmation on the official ETHICS test split. In this repository, the official test is permanently treated as a reused holdout / transfer check.

2 Protocol

The paper package freezes the scientific method around `v9_seed_improvement`. The arm set is: `empty_prompt`, `seed_frozen`, `teacher_single_full_budget_from_seed`, `iterative_from_seed`, and `expert_fixed_benchmark`. The primary comparison is `iterative_from_seed` vs `seed_frozen`. The one-shot full-budget control receives the same total teacher evidence budget as the iterative method, but only one rewrite. The expert prompt is evaluation-only.

The local student is frozen and local-only. The teacher may edit only the structured prompt fields: role summary, core principles, wrong-if bullets, not-wrong-if bullets, anti-heuristics, and tie-break rule. No fine-tuning, retrieval, hidden training, or expert-prompt optimization is allowed.

Nested cross-fitting preserves duplicate groups and isolates `teacher_dev`, `calibration_dev`, `selector_dev`, and `outer_test`. Exploratory publication runs never touch the official test. Compact official runs may launch the reused holdout / transfer check only after the cross-fitted gate passes.

Table 1: Headline publication package outputs.

Artifact	Purpose
`v9_seed_improvement_report.md`	Canonical per-run report
`paper_assets/`	Figures, tables, and summary markdown
`v9_seed_repair_publication.pdf`	Reviewer-facing manuscript PDF
`teacher_manifests/summary.json`	Audit trail for teacher calls
`metadata.json`	Run metadata, phase status, and file map

3 Main Reviewer Path

The canonical reviewer flow is:

1. read the repository README and publication docs;
2. inspect the publication report for the selected run;
3. read Figure 2 and Table 1;
4. inspect the fold-stability appendix assets;
5. verify the teacher-manifest summary and release bundle contents.

4 Mechanism Interpretation

The publication package reports both *signed bias* (`predicted unacceptable rate − gold unacceptable rate`) and *absolute bias magnitude*. This distinction matters because a gain in accuracy can otherwise hide an undesirable drift in label balance. The intended interpretation rule is conservative: if the main gain coincides with strong bias repair but the challenge slice remains weak, the mechanism should be described as calibration or threshold repair with some net correction, not as a broad reasoning breakthrough.

5 Artifact and Release Package

The repository includes a publication-only report generator, asset generator, paper builder, and release exporter. Publication outputs live under `outputs/v9_publication/<run_name>/` and `reports/v9_publication/<run_name>/`. The release bundle lives under `release/v9_seed_improvement/` and contains the report, manuscript PDF, config snapshot, teacher prompt template, paper assets, metadata, and teacher-manifest summary.

6 Limitations

Even with the hardened publication path, the scientific claim is narrow. The study concerns one benchmark, one small local student family, prompt-only interventions, and a teacher that must work through concise structured edits. The compact official Pro run is the reportable endpoint; exploratory Flash runs remain valuable for protocol freeze and sensitivity analysis, but not for the headline claim.

7 Conclusion

The right way to present this project is as a seed-repair study, not a maximalist prompt-optimization claim. That framing makes the core question testable, the controls fair, and the public artifact much stronger. The publication package is designed so a reviewer can audit the exact evidence path from config and manifest through paired statistics, figures, manuscript, and release bundle.

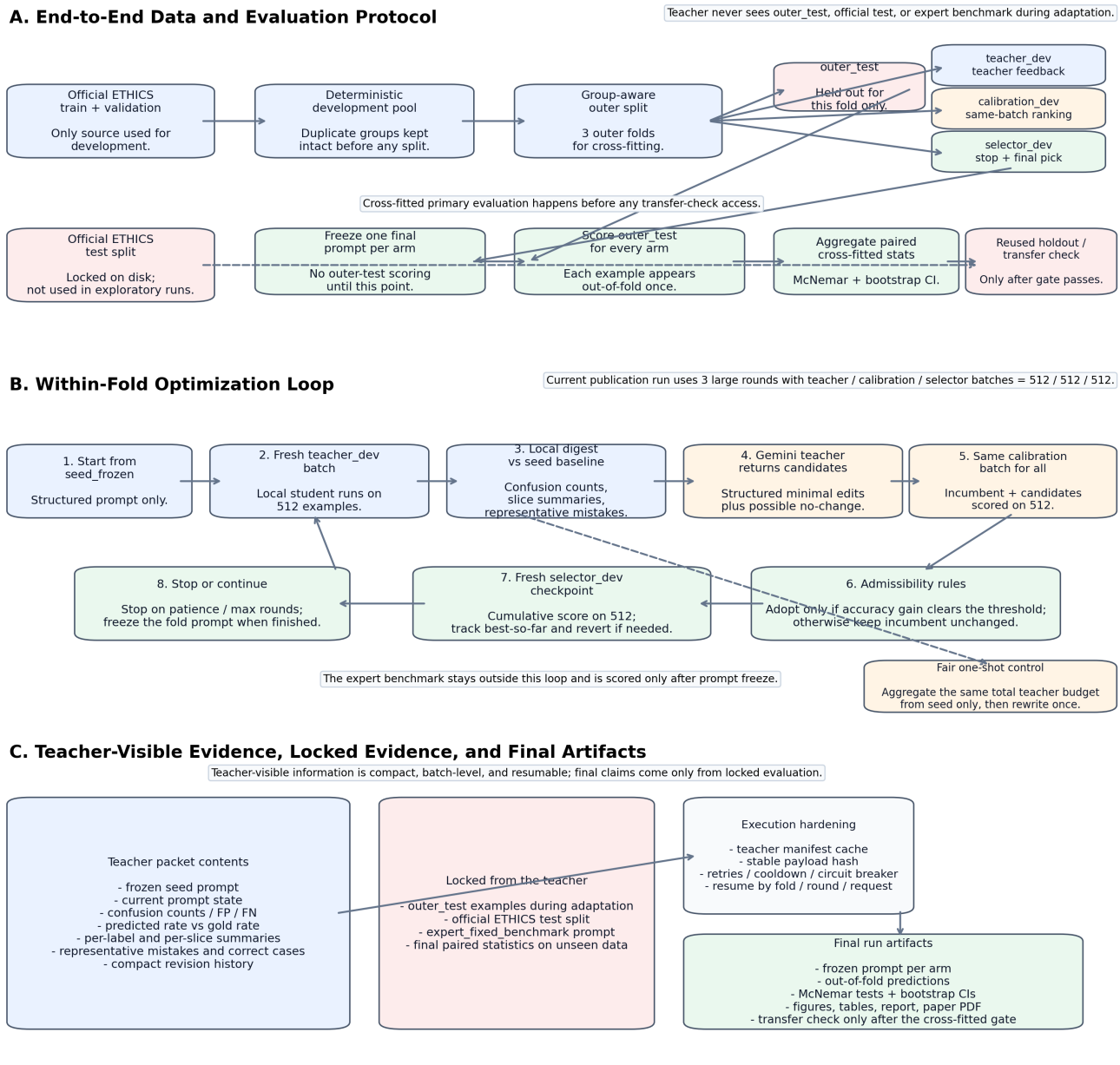

Figure 1: Detailed method maps for the actual publication run. Panel A shows the data protocol from official train + validation through cross-fitted outer-test scoring and the gated reused holdout. Panel B shows the within-fold optimization loop with large shuffled teacher, calibration, and selector batches, same-batch candidate comparison, and the no-change / revert logic. Panel C shows exactly what the teacher can see, what remains locked, and which artifacts are produced at the end.

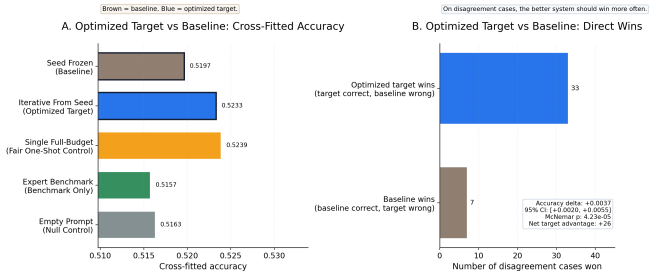

Figure 2: Seed repair at a glance: Panel A shows overall cross-fitted accuracy with the optimized target and baseline highlighted directly, and Panel B shows the primary paired evidence: when the target and baseline disagree, which one is correct more often.

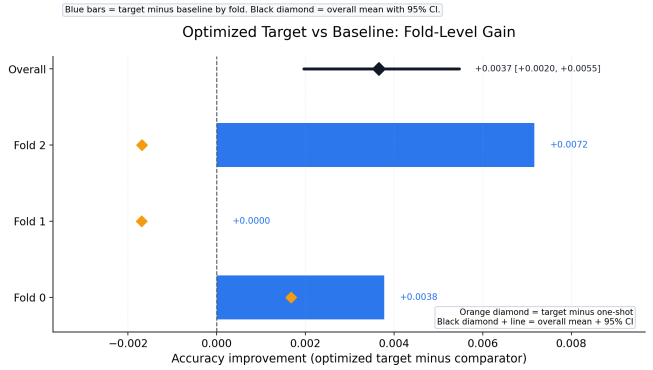

Figure 3: Fold-level stability for `iterative_from_seed` as the optimized target relative to `seed_frozen` as the baseline, plus an overall improvement summary with a 95% confidence interval.

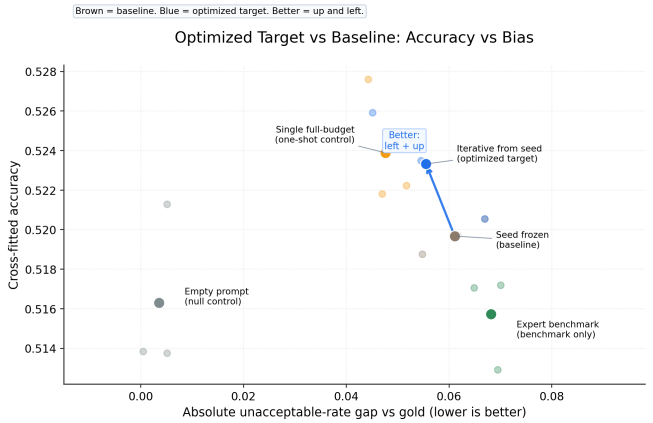

Figure 4: Mechanism view: accuracy versus absolute unacceptable-rate gap. Better models move up and left.